

JAMES RUSE AGRICULTURAL HIGH SCHOOL
YEAR 12 MATHEMATICS EXTENSION I
TRIAL EXAM 2004

QUESTION 1

Marks

- (a) Find $\frac{d}{dx}(\ln(5 + e^x))$ 2
- (b) Find $\int \frac{19 dx}{4 + 8x^2}$ 2
- (c) Evaluate $\int_6^{22} x\sqrt{x+3} dx$ using the substitution $u^2 = x + 3$ 4
- (d) Solve for x : $\frac{x+1}{x-3} \geq 2$ 2
- (e) Six identical yellow discs and four identical blue discs are placed in a straight line.
- (i) How many arrangements are possible ? 1
- (ii) Find the probability that all the blue discs are together. 1

QUESTION 2 (START A NEW PAGE)

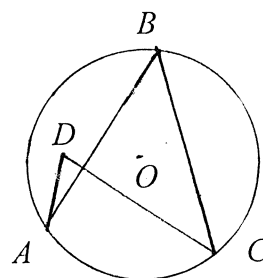
- (a) Find the acute angle (to nearest degree) between the lines :

$$y = \frac{3x}{8} - \frac{7}{8} \quad \text{and} \quad 2x + y - 5 = 0$$

- (b) Points A, B and C lie on the circumference of a circle with centre O , and point D lies inside the circle with

$$\angle ABC = 17^\circ \text{ and } \angle ADC = 34^\circ.$$

Prove $ADOC$ is a cyclic quadrilateral.



(c)

Find $\int \frac{4x-1}{\sqrt{9-x^2}} dx$

(d)

Evaluate $\int_0^1 (1+x^2)^4 dx$

(e)

Find $\frac{d}{dx}(\cos^{-1}(2\cos^2 x - 1))$ in simplest terms for $\{0 \leq x \leq \frac{\pi}{2}\}$.

QUESTION 3 (START A NEW PAGE)

- (a)(i) On the same x - y axes graph the functions $y = f(x)$ and $y = f^{-1}(x)$ if $f(x) = e^x + e^{2x}$. Show all the y intercepts and asymptotes. 3
- (ii) Find the equation of the inverse function $f^{-1}(x)$ if $f(x) = e^x + e^{2x}$ stating the domain and range of $f^{-1}(x)$. 4
- (b) If α is a multiple root of $P(x)=0$ then $P'(\alpha)=0$. 5

Factorise $P(x) = 12x^3 - 16x^2 + 7x - 1$ if $P(x)$ has multiple zeros.

QUESTION 4 (START A NEW PAGE)

- (a) A particle moves in a straight line.

The displacement function x metres in terms of time t seconds is given by :

$$x(t) = 6 \sin 2t - 6 \cos 2t$$

- (i) Show that the displacement function can be written in the form :

2

$$x(t) = R \sin(2t - \alpha) \quad \text{where } R > 0 \text{ and } \{0 < \alpha < 2\pi\}.$$

State the exact values of R and α .

- (ii) Graph the displacement function $x(t)$ for $\{0 < t < 2\pi\}$.

2

- (iii) Show that the motion is Simple Harmonic Motion.

2

- (iv) Find the expression v^2 in terms of displacement x if v is the velocity of the particle.

2

- (v) Find the first time the particle is 2 metres from the centre of motion.

2

- (b) Find the constant term in the expression $x^3 \left(x^2 + \frac{2}{x} \right)^6$

2

QUESTION 5

- (a) A man has a loan of \$ 15800 with monthly reducible interest of 8% p.a.

5

If the repayments are \$1250 per month, find the number of payments to repay all the loan.

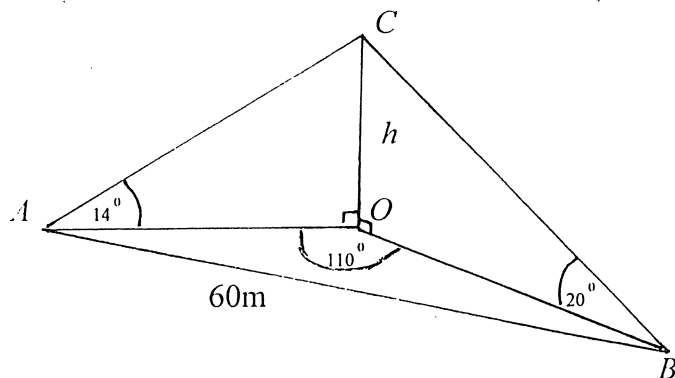
- (b) Prove by induction for all positive integers n :

4

$$\frac{5}{6} + \frac{1}{4} + \dots + \frac{n+4}{n(n+1)(n+2)} = \frac{3}{2} - \frac{n+3}{(n+1)(n+2)}$$

- (c)

3



A vertical tower shown above has angles of elevation from A and B of 14° and 20° respectively.

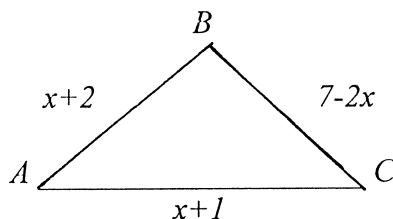
If the distance AB is 60 metres and $\angle AOB = 110^\circ$, find the height h of the tower to the nearest metre.

QUESTION 6

- (a) A bowman fires an arrow with an initial velocity of 50 m/s from 1.5 metres above ground to a target 80 metres away. The bullseye of the target is 0.3 metres in diameter, and the centre of the bullseye is 1 metre above ground.
- (i) Show that the trajectory equation for the flight of the arrow is given by : 3
 $y = x \tan \alpha - \frac{x^2}{500} (1 + \tan^2 \alpha) + 1.5$ where α is the initial angle of elevation of the arrow, the acceleration due to gravity g is 10 m/s^2 and the Origin is at ground level .
- (ii) Find the range of values of α (to the nearest second) for the arrow to hit the bullseye. 5
- (b) The bowman has a probability of $\frac{3}{5}$ of hitting the bullseye.
- (i) Find the probability of hitting the bullseye exactly 7 times from 13 trials. 1
- (ii) By comparing the terms of $\left(\frac{3}{5} + \frac{2}{5}\right)^{13}$ find the most likely outcome of hitting the bullseye from 13 trials. 3

QUESTION 7

- (a) The rate of growth of a population N over t years is given by : $\frac{dN}{dt} = -k(N - 700)$.
- (i) Show $N = 700 + Ae^{-kt}$ satisfies $\frac{dN}{dt} = -k(N - 700)$ where A and k are constants. 1
- (ii) The population has decreased from an initial population of 8300 to 5100 in 5 years. 3
 Find the population at the end of the next 5 years.
- (b) Triangle ABC is shown .



- (i) Show that the domain of x for the triangle to exist is given by $\{ 1 < x < 3 \}$. 2
- (ii) The area A of a triangle with sides a , b and c is given by : 2

$$A = \sqrt{s(s-a)(s-b)(s-c)} \quad \text{where } s = \frac{1}{2}(a+b+c)$$

Show that the expression for the area A of the triangle ABC in terms of x is given by :

$$A = \sqrt{10(x^3 - 8x^2 + 19x - 12)}$$

- (iii) Find the value of x that gives the maximum area of $\triangle ABC$. 4

END OF EXAM

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$$(a) \frac{d}{dx} \ln(5+e^x) = \frac{e^x}{5+e^x}$$

$$(b) \int \frac{19 dx}{4+8x^2} = \frac{19}{8} \int \frac{dx}{x^2 + \frac{1}{2}} = \frac{19}{8} \sqrt{2} \tan^{-1} \sqrt{2}x + C$$

$$(c) \int_6^{22} x \sqrt{x+3} dx$$

$u^2 = x+3$
 $2u du = dx$
 $x=22 \quad u=5$
 $x=6 \quad u=3$

$$\int_3^5 (u^2-3) u \cdot 2u du$$

$$2 \int_3^5 u^2 (u^2-3) du$$

$$2 \int_3^5 u^4 - 3u^2 du$$

$$2 \left[\frac{u^5}{5} - u^3 \right]_3^5$$

$$2 \left[625 - 125 - \left(\frac{243}{5} - 27 \right) \right]$$

$$456 \frac{4}{5}$$

$$(d) \frac{x+1}{x-3} \geq 2 \quad x \neq 3$$

$$(x-3)(x+1) \geq 2(x-3)^2$$

$$(x-3)[x+1-2(x-3)] \geq 0$$

$$(x-3)(-x+7) \geq 0$$

$$(x-3)(x-7) \leq 0$$

$$\text{Soln } \{ 3 < x \leq 7 \}$$

$$(e) (i) \frac{10!}{6! 4!} = 210$$

$$(ii) \text{Probability} = \frac{7}{210} = \frac{1}{30}$$

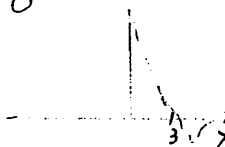
$$(f) m_1 = \frac{3}{8} \quad m_2 = -2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{3}{8} - (-2)}{1 + \frac{3}{8} \cdot (-2)} \right|$$

$$= \left| \frac{19}{2} \right|$$

$$\theta = 84^\circ$$



(k) $\angle AOC = 2\angle ABC$ (Angle at the Centre of a circle is twice the angle at the circumference standing on the same arc.)

$$= 2 \times 17$$

$$= 34^\circ$$

But $\angle AOC = \angle ADC = 34^\circ$

$\therefore$ ADOC is cyclic (If an interval subtends equal angles at two points on the same side of it then the endpoints of the interval and the two points are concyclic.)

$$\Rightarrow \int \frac{4x-1}{\sqrt{9-x^2}} dx = \int \left(4x(9-x^2)^{-\frac{1}{2}} - \frac{1}{\sqrt{9-x^2}} \right) dx$$

$$= -4\sqrt{9-x^2} - \sin^{-1} \frac{x}{3} + C$$

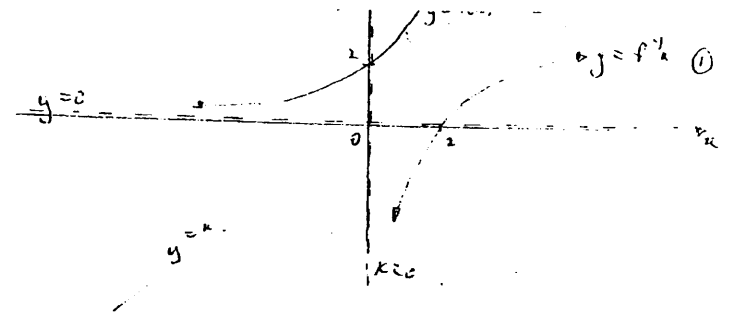
$$\therefore \int_0^1 (1+x^2)^4 dx = \int_0^1 (1 + 4x^2 + 6x^4 + 4x^6 + x^8) dx$$

$$= \left[x + \frac{4}{3}x^3 + \frac{6}{5}x^5 + \frac{4}{7}x^7 + \frac{x^9}{9} \right]_0^1$$

$$= 4 \frac{68}{315}$$

$$(e) \frac{d}{dx} \cos^4(2\cos^2 x - 1) = \frac{d}{dx} \cos^4(\cos 2x) \quad \left\{ 0 < x < \frac{\pi}{2} \right\}$$

$$= \frac{d}{dx} 2x$$



(i)

$$x = e^y + e^{2y}$$

$$(e^y)^2 + e^y - x = 0$$

$$e^y = \frac{-1 \pm \sqrt{1+4x}}{2}$$

$$y = \ln \left[\frac{\sqrt{1+4x} - 1}{2} \right] \text{ Only } \text{ as } e^y > 0 \quad (1)$$

$$\text{Domain } \{x > 0\}$$

$$\text{Range : defined for all } y \quad (1)$$

(ii)

$$P(x) = 12x^3 - 16x^2 + 7x - 1$$

$$P'(x) = 36x^2 - 32x + 7 \quad (1)$$

$$P'(x) = 0$$

$$36x^2 - 32x + 7 = 0$$

$$(2x-1)(18x-7) = 0$$

$$x = \frac{1}{2} \text{ or } x = \frac{7}{18}$$

$$P\left(\frac{1}{2}\right) = 12\left(\frac{1}{2}\right)^3 - 16\left(\frac{1}{2}\right)^2 + \frac{7}{2} - 1$$

$$= 0 \quad \text{as } 2x-1 \text{ is a factor of } P(x) \quad (1)$$

$$\therefore P(x) = (2x-1)^2 Q(x)$$

$$= (2x-1)^2 (3x-1) \quad (1)$$

$$(i) \quad x(t) = 6 \sin 2t - 6 \cos 2t$$

$$R \sin(2t - \frac{\pi}{4}) = R \cos 2t \sin \frac{\pi}{4} - R \sin 2t \cos \frac{\pi}{4}$$

$$\therefore R \sin 1 = 6$$

$$R \cos 1 = 6$$

$$R \sin 20 \quad \text{and} \quad R \cos 20 \quad \left. \vphantom{\begin{matrix} R \sin 20 \\ R \cos 20 \end{matrix}} \right\} \left\{ \begin{matrix} 0 < 20 < \frac{\pi}{2} \end{matrix} \right\}$$

$$R = \sqrt{6^2 + 6^2}$$

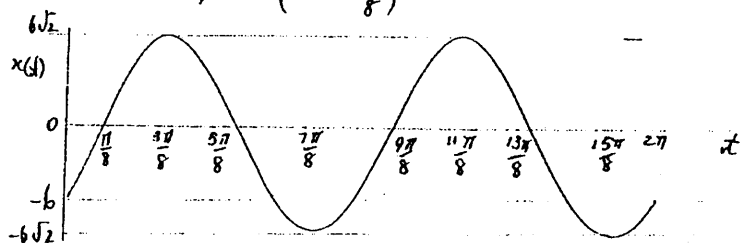
$$= 6\sqrt{2}$$

$$\tan \phi = 1$$

$$\phi = \frac{\pi}{4}$$

$$\therefore x(t) = 6\sqrt{2} \sin(2t - \frac{\pi}{4})$$

$$= 6\sqrt{2} \sin 2(t - \frac{\pi}{8})$$



$$\dot{x}(t) = 12\sqrt{2} \cos(2t - \frac{\pi}{4})$$

$$\ddot{x}(t) = -24\sqrt{2} \sin(2t - \frac{\pi}{4})$$

$$= -4 [6\sqrt{2} \sin(2t - \frac{\pi}{4})]$$

$$\ddot{x} = -4x$$

which is of the form $\ddot{x} = -n^2(x - s)$

$$\therefore \text{motion SHM } n=2 \quad s=0$$

$$v^2 = 288 \cos^2(2t - \frac{\pi}{4})$$

$$= 288 (1 - \sin^2(2t - \frac{\pi}{4}))$$

$$= 288 (1 - (\frac{x}{6\sqrt{2}})^2)$$

$$v^2 = 288 - 4x^2$$

$$(v) \quad -2 = 6\sqrt{2} \sin(2t - \frac{\pi}{4})$$

$$\sin(2t - \frac{\pi}{4}) = -\frac{1}{3\sqrt{2}}$$

$$2t - \frac{\pi}{4} = -0.24$$

$$t = \frac{1}{2} \left[\frac{\pi}{4} - 0.24 \right]$$

$$t = 0.27 \text{ seconds.}$$

$$(b) \quad x^3 \left[x^2 + \frac{2}{x} \right]^6$$

$$T_{r+1} = {}^6C_r (x^2)^{6-r} \left(\frac{2}{x} \right)^r$$

$$= {}^6C_r x^{3+12-2r-r}$$

$$= {}^6C_r 2^r x^{15-3r}$$

Constant term $r=5$.

$$\therefore T_6 = {}^6C_5 2^5$$

$$= 192$$

$$\frac{5}{(a)} \quad \text{monthly interest} = \frac{8}{1200} = \frac{1}{150}$$

$$\text{Amount owing end 1st Payment} = 15800 \left[1 + \frac{1}{150} \right] - 1250$$

$$= 15800 \cdot \frac{151}{150} - 1250$$

$$\text{Amount owing end 2nd Payment} = \left[15800 \left(\frac{151}{150} \right) - 1250 \right] \frac{151}{150} - 1250$$

$$= 15800 \left(\frac{151}{150} \right)^2 - 1250 \left[1 + \frac{151}{150} \right]$$

$$\text{Amount owing end 3rd Payment} = \left[15800 \left(\frac{151}{150} \right)^2 - 1250 \left(1 + \frac{151}{150} \right) \right] \frac{151}{150} - 1250$$

$$0 = 15800 \left(\frac{151}{150} \right)^n - 1250 \left[1 + \frac{151}{150} + \left(\frac{151}{150} \right)^2 + \dots + \left(\frac{151}{150} \right)^{n-1} \right]$$

$$15800 \left(\frac{151}{150} \right)^n = 1250 \left[\frac{\left(\frac{151}{150} \right)^n - 1}{\frac{151}{150} - 1} \right]$$

$$= 187500 \left[\left(\frac{151}{150} \right)^n - 1 \right]$$

$$\therefore (187500 - 15800) \left(\frac{151}{150} \right)^n = 187500$$

$$\left(\frac{151}{150} \right)^n = \frac{187500}{171700}$$

$$n = \frac{\ln \left(\frac{187500}{171700} \right)}{\ln \left(\frac{151}{150} \right)}$$

$$n = 13.24$$

$$\therefore n = 14 \text{ payments}$$

$$b) \frac{5}{6} + \frac{1}{4} + \dots + \frac{1+4}{n(n+1)(n+2)} = \frac{3}{2} - \frac{n+3}{(n+1)(n+2)}$$

$$\text{Step 1 } n=1 \quad S_1 = \frac{3}{2} - \frac{4}{2 \cdot 3} = \frac{9-4}{6} = \frac{5}{6}$$

$$\therefore T_1 = S_1$$

$\therefore$ True for $n=1$

①

$$\therefore \frac{5}{6} + \frac{1}{4} + \dots + \frac{k+4}{k(k+1)(k+2)} = \frac{3}{2} - \frac{k+3}{(k+1)(k+2)}$$

to prove statement is true for $n=k+1$

$$\therefore \frac{5}{6} + \frac{1}{4} + \dots + \frac{k+5}{(k+1)(k+2)(k+3)} = \frac{3}{2} - \frac{k+4}{(k+2)(k+3)}$$

new

$$\frac{5}{6} + \frac{1}{4} + \dots + \frac{k+4}{k(k+1)(k+2)} + \frac{k+5}{(k+1)(k+2)(k+3)}$$

$$= \frac{3}{2} - \frac{k+3}{(k+1)(k+2)} + \frac{k+5}{(k+1)(k+2)(k+3)} \quad \text{By assumption}$$

$$= \frac{3}{2} + \frac{k+5 - (k+3)^2}{(k+1)(k+2)(k+3)}$$

$$= \frac{3}{2} + \frac{-k^2 - 5k - 4}{(k+1)(k+2)(k+3)}$$

$$= \frac{3}{2} - \frac{(k+1)(k+4)}{(k+1)(k+2)(k+3)}$$

$$= \frac{3}{2} - \frac{k+4}{(k+2)(k+3)}$$

①

$\therefore$ If statement is true for $n=k$ it is also true for $n=k+1$

Steps Since statement is true for $n=1$ it also true for $n=1+1=2$, $n=2+1=3$, and so on for all positive integers n .

$$\alpha = \frac{700 - \sqrt{160000 - 4 \times 64 \times 60.15}}{128}$$

$$\angle = 80.681^\circ \text{ or } 8.853^\circ$$

$$= 80^\circ 40' 53'' \quad 8^\circ 51' 11''$$

$$y = 1.15$$

$$1.15 = 80 \tan \alpha - \frac{64}{5} (1 + \tan^2 \alpha) + 1.5$$

$$64 \tan^2 \alpha - 400 \tan \alpha + 67.25 = 0$$

$$\tan \alpha = \frac{400 \pm \sqrt{400^2 - 4 \times 64 \times 67.25}}{128}$$

$$\angle = 80.675^\circ \text{ or } 9.074^\circ$$

$$= 80^\circ 40' 30'' \quad 9^\circ 4' 26''$$

$$\therefore \text{range } \{8^\circ 51' 11'' < \angle < 9^\circ 4' 26''\}$$

$$\text{OR } \{80^\circ 40' 30'' < \angle < 80^\circ 40' 53''\}$$

$$= \frac{1716 \cdot 3^7 \cdot 2^5}{5^{13}}$$

$$\text{OR } 0.20$$

$$(ii) \quad T_{n+1} = \binom{13}{r} p^{13-r} q^r \quad \text{for } (p+q)^{13}$$

$$\frac{T_{n+1}}{T_r} = \frac{\binom{13}{r} p^{13-r} q^r}{\binom{13}{r-1} p^{14-r} q^{r-1}}$$

$$= \frac{13!}{r!(13-r)!} \cdot \frac{(r-1)! (14-r)!}{13!} \cdot \frac{q}{p}$$

$$= \frac{14-r}{r} \cdot \frac{q}{p}$$

$$= \frac{2(14-r)}{3r}$$

$$\frac{T_{n+1}}{T_r} > 1$$

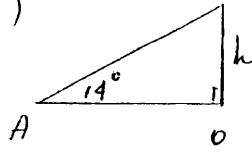
$$\therefore \frac{2(14-r)}{3r} > 1$$

$$5r < 28$$

$$r < 5.6$$

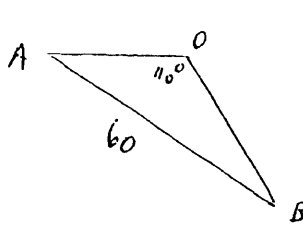
$\therefore$ most likely $r=5 \Rightarrow$ 8 times from 13 to hit bullseye.

(C)



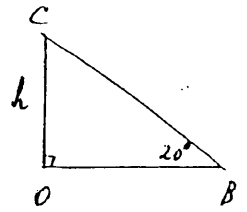
$$\text{Now } \frac{h}{AO} = \tan 14^\circ$$

$$AO = \frac{h}{\tan 14^\circ}$$



$$\text{And } \frac{h}{OB} = \tan 20^\circ$$

$$OB = \frac{h}{\tan 20^\circ}$$



$$\text{But } AB^2 = AO^2 + BO^2 - 2AO \cdot BO \cos 110^\circ \quad (\text{cosine rule})$$

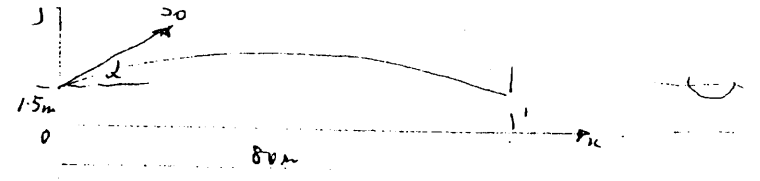
$$60^2 = \frac{h^2}{\tan^2 14^\circ} + \frac{h^2}{\tan^2 20^\circ} - \frac{2h^2 \cos 110^\circ}{\tan 14^\circ \tan 20^\circ}$$

$$h^2 = \frac{60^2 \tan^2 14^\circ \tan^2 20^\circ}{\tan^2 14^\circ + \tan^2 20^\circ - 2 \cos 110^\circ \tan 14^\circ \tan 20^\circ}$$

$$h = 10.75$$

∴ height tower = 11m (nearest m).

E



$$m \ddot{y} = -mg$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$t = 0 \quad \dot{y} = 50 \text{ m/s} \Rightarrow C = 50 \text{ m/s}$$

$$\therefore \dot{y} = -10t + 50 \text{ m/s}$$

$$y = -5t^2 + 50t \text{ m/s} + c$$

$$t = 0 \quad y = 1.5 \Rightarrow C = 0$$

$$\therefore y = -5t^2 + 50t \text{ m/s} + 1.5$$

For trajectory

$$y = -5 \left[\frac{x}{50 \cos 2^\circ} \right]^2 + 50 \cos 2^\circ \cdot \frac{x}{50 \cos 2^\circ} + 1.5$$

$$= -\frac{5}{2500} x^2 \sec^2 2^\circ + x \tan 2^\circ + 1.5$$

$$y = x \tan 2^\circ - \frac{x^2}{500} (1 + \tan^2 2^\circ) + 1.5$$

$$(ii) \quad \text{Bullseye is } \left\{ 1 - \frac{0.3}{2} \leq y \leq 1 + \frac{0.3}{2} \right\}$$

$$\left\{ 0.85 \leq y \leq 1.15 \right\}$$

$$\therefore y = 0.85 \quad x = 80$$

$$\therefore 0.85 = 80 \tan 2^\circ - \frac{64}{5} (1 + \tan^2 2^\circ) + 1.5$$

$$\frac{64}{5} \tan^2 2^\circ - 80 \tan 2^\circ + \frac{64}{5} - 0.65 = 0$$

$$64 \tan^2 2^\circ - 400 \tan 2^\circ + 60.75 = 0$$

$$(i) N = 700 + Ae^{-kt}$$

$$\frac{dN}{dt} = -Ake^{-kt}$$

$$\text{But } Ae^{kt} = N - 700.$$

$$\therefore \frac{dN}{dt} = -k[N - 700]$$

$$(ii) t = 0 \quad N = 8300$$

$$\therefore 8300 = 700 + A$$

$$A = 7600.$$

$$\therefore N = 700 + 7600e^{-kt}$$

$$t = 5 \quad N = 5100$$

$$\therefore 5100 = 700 + 7600e^{-5k}$$

$$k = \frac{1}{5} \ln \frac{7600}{4400}$$

$$\text{or } k = \frac{1}{5} \ln \left(\frac{19}{11} \right)$$

$$t = 10 \quad N = 700 + 7600e^{-2 \ln \left(\frac{19}{11} \right)}$$

$$N = 3247$$

(i) Given two sides triangle > third side

$$(x+1) + x+2 > 7-2x \quad x+2 + 7-2x > x+1 \quad \text{or } x+1 + 7-2x > x+2$$

$$4x > 4$$

$$-2x > -8$$

$$-2x > -6$$

$$x > 1$$

$$x < 4$$

$$x < 3.$$

$$\therefore \text{Domain } \{1 < x < 3\}$$

$$A = \frac{1}{2} [x+1 + x+2 + 7-2x]$$

$$= \frac{1}{2} \cdot 10$$

$$= 5$$

$$\therefore A = \sqrt{5(5-(x+1))(5-(x+2))(5-(7-2x))}$$

$$= \sqrt{5(4-x)(3-x)(2x-2)}$$

$$= \sqrt{10(x^2-7x+12)(x-1)}$$

$$A = \sqrt{10(x^3-8x^2+19x-12)}$$

$$(iii) \frac{dA}{dx} = \frac{\frac{1}{2} \cdot 10 [3x^2-16x+19]}{\sqrt{10(x^3-8x^2+19x-12)}} \\ = \frac{5 [3x^2-16x+19]}{\sqrt{10(x^3-8x^2+19x-12)}} \quad (1)$$

$$\text{For maximum area } \frac{dA}{dx} = 0$$

$$\therefore 3x^2-16x+19 = 0$$

$$x = \frac{16 \pm \sqrt{256-4 \cdot 3 \cdot 19}}{6}$$

$$= \frac{16 \pm \sqrt{28}}{6}$$

$$= \frac{8-\sqrt{7}}{3} \quad \text{or} \quad \frac{8+\sqrt{7}}{3} \quad (1)$$

$$\text{But domain } x \text{ is } \{1 < x < 3\}$$

$$\therefore x = \frac{8-\sqrt{7}}{3} \text{ only.}$$

For nature of turning point test $\frac{d^2A}{dx^2}$ since

A and $\frac{dA}{dx}$ are continuous in domain $\{1 < x < 3\}$

A	1.7	$\frac{8-\sqrt{7}}{3}$	1.8
$\frac{dA}{dx}$	0.51	0	-0.09

/ — \

①

$\therefore$ there is an maximum at $x = \frac{8-\sqrt{7}}{3}$

but since there is only one turning point in the domain $\{1 < x < 3\}$ then $x = \frac{8-\sqrt{7}}{3}$ is an absolute maximum.